Part 1 – Question Bank

10-Mark Questions

1. Explain the concept of **Web Mining** and discuss its three primary categories in detail.
2. Describe the **architecture and functional modules of a modern Search Engine**.
3. Discuss the **Hyperlink-Induced Topic Search (HITS) algorithm**, including its mathematical working.
4. Explain the **PageRank algorithm**, its probabilistic foundation, and provide a calculated example.
5. Analyze the various types of **complex data mining**, focusing specifically on **sequence and graph data**.
6. Discuss the major applications of **Data Mining in the Finance and Retail industries**.

5-Mark Questions

7. Explain the common techniques used in **Web Content Mining**.
8. Discuss **Web Usage Mining** and the important parameters collected for analysis.
9. Compare **Content-Based Filtering** and **Collaborative Filtering** in Recommender Systems.
10. Define **Precision and Recall** and explain how they evaluate search engine quality.
11. Examine the methods used for **Privacy-Preserving Data Mining**.
12. Describe the classification and analysis of **Time-Series Data**.
13. Explain the mining of **Spatiotemporal** and **Cyber-Physical System (CPS)** data.
14. Discuss the current trends in **Big Data Mining** and its integration with **AI**.

2-Mark Questions

15. Define the **“Scarcity”** and **“Abundance”** problems in web content mining.
16. What is a **Web Crawler**, and what are its alternate names?
17. Define **“Shingles”** in the context of document fingerprinting.
18. Distinguish between **“Stop Words”** and **“Stemming”** in indexing.
19. What is a **“Damping Factor”** in the PageRank formula?
20. Define **Spatial Data Mining**.
21. What is the difference between a **Hub Page** and an **Authority Page**?
22. Define **Sequence Classification**.
23. What is the difference between **Homogeneous** and **Heterogeneous networks**?
24. Define **Multimedia Data Mining**.

Part 2 – Answer Bank

10-Mark Questions

1. Explain the concept of Web Mining and its three primary categories in detail.

Web mining is the specialized application of data mining and machine learning techniques to extract useful patterns and knowledge from the vast resources available on web pages.

- **Web Content Mining Focus:** This category concentrates on extracting valuable information from the actual data contained within web pages, such as text, images, and videos. It analyzes the internal content while link structures and user interactions are ignored during this specific process.
- **Semi-Structured Challenges:** Most web content is stored in semi-structured HTML formats, which leads to information being scattered across multiple pages rather than in a single location. This complexity requires advanced mining techniques to effectively consolidate data from diverse web sources.
- **Search Result Problems:** Simple keyword-based content mining often encounters the "scarcity problem," where too few results are found, or the "abundance problem," where a query returns an overwhelming number of results. These issues highlight the need for more sophisticated mining algorithms beyond basic frequency counts.
- **Web Structure Mining Overview:** This area focuses on extracting information from the hyperlinks that connect different web pages across the internet. The analysis of these link structures is fundamental for ranking pages according to their perceived importance and relevance.
- **Role of Hyperlinks:** Links between documents serve as a roadmap of the web, allowing mining algorithms to determine how different websites are interconnected. This structural data helps companies and researchers understand the relationships between various commercial or informational domains.
- **Authority Page Identification:** Structure mining helps identify "authority" pages, which are websites that provide reliable and critically important information on a specific topic. These pages are often characterized by having a high number of incoming links from other reputable sources.
- **Hub Page Recognition:** This category also identifies "hub" pages, which act as high-quality directories by pointing to many different authority pages. The core concept is that a good hub points to many authorities, and a good authority is linked by many hubs.
- **Web Usage Mining Basics:** Usage mining involves extracting information from log data generated by users as they interact with different web pages. It primarily uses web server logs, application server logs, and application-level logs to track visitor activities.
- **Predicting User Behaviour:** The primary goal of usage mining is to analyze and predict how users browse websites to improve business strategies and make sites more customer-centric. For example, companies might reallocate advertising budgets if logs show more traffic coming from one social platform over another.
- **Analytical Techniques:** Usage data is typically processed using Association Rule Mining to find relationships between pages visited, such as users who visit page A often visiting page B. Cluster analysis is also used to group users with similar browsing patterns for better personalization.

2. Describe the architecture and working mechanism of a modern Search Engine.

A modern search engine is a complex information retrieval system designed to automatically find and rank relevant web data in response to specific user queries.

- **Web Crawler Function:** The crawler, also known as a spider or bot, is a software component that systematically visits web pages across the internet to fetch data. It acts as the primary data gathering tool that fuels the search engine's database.
- **Parallel Data Retrieval:** Because there are billions of web pages, crawlers must run in parallel on large server clusters to ensure the index remains updated. The data they fetch is temporarily stored so it can be processed later into a searchable format.
- **Indexer Module Role:** The indexer identifies keywords from the fetched pages to create a comprehensive index used for rapid searching. It effectively maps specific keywords to the unique web page IDs where those words were found.
- **HTML Element Extraction:** Keywords are specifically extracted from critical HTML tags like title tags, meta tags, and heading tags to determine the page's core subject. This targeted extraction ensures that the most relevant terms are used for the final index.
- **Stop Word Removal:** During indexing, common words like "the," "and," and "it" are removed because they do not provide distinct meaning to a search query. This process reduces the size of the index and improves processing efficiency.
- **Word Stemming:** Stemming is a preprocessing step that converts different variations of a word into its single root form, such as converting "computing" to "computer". This allows the search engine to match a user's query with relevant variations of a term.
- **Query Processor Importance:** The query processor is the most critical part of the system, acting as the interface between the user's request and the stored index. In modern systems, it uses NLP to understand the context and intent behind a user's words.
- **Personalized Result Generation:** Modern processors use user profiles—including location, gender, age, and device type—to provide personalized search results. For example, a search for "hospitals" in Delhi will prioritize nearby facilities over those in other cities.
- **Query Checking and Categorization:** Before execution, the processor checks for spelling mistakes and categorizes the query type, such as news, products, or services. This categorization helps the system select the most effective strategy for retrieving results from the index.
- **Ranking Results (SERP):** The final step involves arranging the retrieved pages by relevance on a Search Engine Results Page (SERP) using algorithms like PageRank. This ensures that the most authoritative and useful content appears at the top for the user.

3. Discuss the Hyperlink-Induced Topic Search (HITS) algorithm, including its mathematical approach.

The HITS algorithm, developed by Jon Kleinberg in 1999, is a link analysis method used to rank web pages based on the concepts of hubs and authorities.

- **Authority Page Concept:** Authority pages are those that contain useful, reliable, and high-quality information on a particular topic. These are the pages that a user is typically searching for when they enter a query.
- **Hub Page Concept:** Hub pages act as high-quality directories or portals that provide links to many relevant authority pages. A strong hub doesn't necessarily contain the information itself but points the user to where it can be found.

- **Mutually Reinforcing Relationship:** The HITS algorithm is built on the idea that a good hub points to many good authorities, and a good authority is linked by many good hubs. This creates a circular, reinforcing relationship between the two types of pages.
- **Step 1 - Adjacency Matrix:** The first mathematical step is to represent the web structure as an adjacency matrix (L), where rows and columns represent pages and entries show the presence of links. If page 1 links to page 2, the entry at is set to 1, otherwise it is 0.
- **Step 2 - Matrix Transpose:** To calculate authority scores, the transpose of the link matrix () is required. The transpose flips the matrix, effectively turning outgoing links into incoming links for mathematical processing.
- **Step 3 - Initial Authority Weight:** Initially, every web page is assigned an authority weight vector value of 1. This serves as the starting point for the first iteration of the ranking calculation.
- **Step 4 - Hub Score Calculation:** The hub weight vector (H) is calculated by multiplying the original link matrix (L) by the authority weight vector (A). This sums up the authority scores of all pages that a particular page points to.
- **Step 5 - Authority Score Calculation:** The authority weight vector (A) is then calculated or updated by multiplying the transpose matrix () by the hub vector (H). This counts the hub scores of all pages that link into a particular page.
- **Iterative Refinement:** These steps can be repeated multiple times to obtain increasingly accurate weights for both hubs and authorities. In each iteration, the scores are refined based on the updated values of the other vector.
- **Final Ranking Interpretation:** After the iterations, pages with the highest authority weights are identified as the most relevant sources for a keyword. For instance, a page hyperlinked to many high-ranking hubs becomes a strong authority.

4. Explain the PageRank algorithm, its probabilistic foundation, and provide a calculated example.

PageRank is the foundational link-analysis algorithm used by Google to rank web pages by measuring their relative importance based on backlinks.

- **Backlink Importance:** The core philosophy of PageRank is that a web page is considered important if many other pages link to it. This is analogous to social behaviour, where an individual is viewed as important if they are known by many other important people.
- **Probabilistic Foundation:** The algorithm is rooted in probability theory, estimating the likelihood that a "random surfer" will click on a particular link. It assumes that a user on a page is equally likely to click any link present on that page.
- **Inversion Proportion Rule:** The probability of clicking a specific link is inversely proportional to the total number of links on the current page. For example, if page A has two links, there is a 50% chance of a user following either one.
- **The Damping Factor ():** The PageRank formula includes a damping factor (), which represents the probability that the user will continue clicking links. A standard value for is often around 0.85, reflecting realistic browsing habits.
- **Formula Components:** The PageRank of page A is calculated as , where is the number of outgoing links from page T. This balances the chance of arriving via a link versus arriving directly.
- **User Exit Scenario:** The term in the formula specifically accounts for the scenario where a user gets bored and leaves the web page or the site entirely. This ensures the ranking doesn't become stuck in infinite link loops.

- **Example - Step 1:** Suppose there are three web pages, A, B, and C, and all are assigned an initial PageRank of 1. This initial equal distribution is the starting point for the ranking flow.
- **Example - Step 2:** If page A links to both B and C, it distributes its rank by giving 0.5 to each. If page B links only to C, it gives its full rank of 1 to C. If C links only to A, it gives its full rank of 1 back to A.
- **Example - Step 3:** After one iteration, page C receives a rank from both A (0.5) and B (1), while A receives rank only from C (1), and B receives rank only from A (0.5). This makes C the highest-ranked page because it has the most incoming link weight.
- **Interpretation of Results:** In this example, C is most important due to receiving the most incoming "votes," followed by A, with B being the least important. This simple flow demonstrates how PageRank successfully identifies relative importance through link structure.

5. Analyze the various types of complex data mining, focusing on sequence and graph data.

Complex data mining involves extracting patterns from non-traditional datasets such as ordered sequences and multi-layered networks.

- **Sequence Data Definition:** A sequence is defined as an ordered list of events, which can be further categorized based on the characteristics of the events they describe. These are often long and carry hidden semantic meanings.
- **Time-Series Data Characteristics:** This type of sequence consists of numeric data recorded at equal time intervals, such as stock market prices tracked every minute. Mining here involves similarity searches and trend analysis to predict future movements.
- **Symbolic Sequence Data:** Unlike time-series, symbolic sequences consist of event or nominal data that are not typically observed at equal intervals. Examples include customer shopping sequences and web click streams.
- **Biological Sequences:** These include DNA and protein sequences, which are typically very long and crucial for modern bioinformatics. Analysis focuses on sequence alignment, which assumes that evolutionary closeness results in higher similarity.
- **Sequential Pattern Mining:** This involves finding frequent patterns in symbolic sequences, often using user-specified constraints to narrow the search space. This is known as constraint-based sequential pattern mining.
- **Graph and Network Mining Importance:** With the rise of the web and social media, graph mining has become essential for analyzing complex relationships between objects. It is used in fields ranging from chemical informatics to multimedia retrieval.
- **Homogeneous Networks:** A network is considered homogeneous if all nodes and links are of the same type, such as a simple friend network or a coauthor network. Mining these focuses on clustering and statistical modelling of uniform relationships.
- **Heterogeneous Networks:** These networks contain different types of nodes and links, such as health-care networks linking doctors, patients, diseases, and treatments. Analyzing these requires more complex ranking and classification methods.
- **Graph Pattern Mining Tasks:** This specific task involves mining frequent subgraphs within a single large graph or a set of graphs. Methods are generally categorized into Apriori-based or pattern growth-based approaches.
- **Advanced Graph Applications:** Graph mining also includes role discovery, link prediction in information networks, and the study of how these networks evolve over time. This helps in identifying key influencers or predicting future connections in a social structure.

6. Discuss the major applications of Data Mining in industries like Finance, Retail, and Science.

Data mining is increasingly integrated across diverse sectors to improve decision-making, efficiency, and discovery through the analysis of massive datasets.

- **Financial Loan Prediction:** Banks use data mining for loan payment prediction and customer credit policy analysis. This helps financial institutions assess the risk of lending to specific individuals based on historical data.
- **Targeted Financial Marketing:** By using classification and clustering, banks can group customers for targeted marketing campaigns. This ensures that financial products are offered to the most relevant and likely interested audience.
- **Financial Crime Detection:** Data mining is a critical tool for detecting money laundering and other financial crimes. It identifies unusual patterns in transactions that may indicate fraudulent or illegal activities.
- **Retail Buying Behaviour:** In the retail industry, mining helps identify customer buying behaviours and shopping patterns. This information allows businesses to adjust their stock and marketing strategies according to current trends.
- **Retail Customer Retention:** Data mining analyzes customer loyalty to help businesses improve retention and satisfaction rates. By understanding why customers stay or leave, retailers can design better loyalty programs.
- **Retail Product Recommendation:** Recommender systems in retail suggest products to users based on cross-referencing items other customers have purchased. This cross-selling strategy enhances goods consumption ratios and boosts sales.
- **Scientific Data Warehousing:** Scientific domains like astronomy and meteorology collect vast amounts of data using telescopes and satellite sensors. Data mining provides the necessary preprocessing and warehousing to make this data searchable and useful.
- **Mining Complex Scientific Data:** Science and engineering rely on mining complex data types, including graph-based and network-based mining. This is essential for fields like geosciences and biological pathway analysis.
- **Scientific Visualization Tools:** Data mining in science often integrates with visualization tools to help researchers see patterns in huge datasets. This allows for better human integration with automated discovery processes.
- **Engineering System Monitoring:** In computer science and engineering, data mining monitors system status and isolates software bugs. It is also used to recognize system malfunctions and uncover network intrusions in real-time.

5-Mark Questions

7. Explain the common techniques used in Web Content Mining.

Web content mining uses various analytical approaches to manage and extract information from the massive number of documents available online.

- **Web Document Clustering:** This approach manages large numbers of documents by forming meaningful clusters based on keywords rather than just a ranked list. Techniques like K-means and agglomerative clustering are commonly used to group similar pages together.
- **Resemblance Measurement:** Resemblance is used to improve query results by identifying and removing duplicate or nearly identical web pages. It calculates a value between 0 and 1, where 1 indicates that two documents are exactly the same.

- **Containment Measurement:** Containment measures whether one document exists entirely inside another document. A value of 1 means the first document is fully contained in the second, helping to filter redundant partial content.
- **Document Fingerprinting:** Fingerprinting is used to measure similarity by dividing a document into a continuous sequence of fixed-length words called "shingles". The similarity is determined by counting how many shingles match between two different documents.
- **Shingle Analysis:** Using shingles (for example, groups of 2 words) allows for very accurate similarity detection. While accurate, this method can be inefficient when applied to extremely large datasets across the web.

8. Discuss Web Usage Mining and the important parameters collected for analysis.

Web usage mining extracts patterns from user interaction logs to predict behaviour and optimize website design.

- **Visitor and Hit Counts:** Mining collects the total number of hits (requests) and the total number of unique visitors within a specific time duration. This helps site owners understand the basic volume of traffic their site receives.
- **Referrer and Landing Pages:** The "referrer" parameter identifies the website the user came from, while the "landing page" is the specific entry point where they first arrived. This is crucial for evaluating the effectiveness of external marketing or social media campaigns.
- **Session and Path Analysis:** Parameters include session duration (time spent on site) and path analysis (the sequence of pages visited). These metrics help in restructuring websites to improve the overall user experience and find-ability.
- **Technical Environment Details:** Logs record the browser type, device type (mobile/desktop), and operating system used by the visitor. This data allows developers to optimize the website for the most popular hardware and software combinations.
- **Cookies and Authentication:** Cookies are small amounts of data stored in the user's browser to identify them and save authentication tokens. They are essential for personalized usage mining and maintaining user state across sessions.

9. Compare Content-Based Filtering and Collaborative Filtering in Recommender Systems.

Recommender systems suggest items to users based on two primary data filtering methodologies.

Feature	Content-Based Filtering	Collaborative Filtering
Data Basis	Based on a **single user's** specific interactions and preferences.	Based on interactions and data collected from **many users**.
Metadata Use	Uses metadata from the user's own history, like past likes or views.	Casts a wider net to find patterns across the entire user base.
Logic	Suggests products/services that are **similar** to what the user liked before.	Suggests items based on other users with **similar tastes** or situations.
Example	Recommending a blue dress because the user bought a blue dress.	Recommending a dress because a "similar customer" liked it.
Personalization	Highly focused on individual history and specific item attributes.	Focused on communal trends and shared interests between people.

10. Define Precision and Recall and explain how they evaluate search engine quality.

Precision and recall are the two primary metrics used to evaluate the effectiveness and quality of a search engine's results.

- **Precision Definition:** Precision measures how many of the results retrieved by the search engine are actually relevant to the user's query. It is the ratio of relevant pages retrieved to the total number of pages retrieved.
- **Recall Definition:** Recall measures how many of the total relevant results available on the internet were successfully retrieved by the engine. It is the ratio of relevant pages retrieved to the total relevant pages available.
- **Calculation Example:** If there are 200 relevant pages on the web and a search retrieves 190 of them (out of 210 total results), the precision is 90.48%. In this same example, the recall is 95%, indicating a very comprehensive search.
- **Venn Diagram Context:** These measures are often visualised using a Venn diagram where Circle A is retrieved results and Circle B is relevant results. The intersection represents the "Relevant Retrieved" items used in both formulas.
- **Evaluation Goal:** A high-quality search engine aims for high scores in both categories, meaning it provides accurate results (precision) and doesn't miss important information (recall).

11. Examine the methods used for Privacy-Preserving Data Mining.

Privacy-preserving data mining seeks to obtain valid analytical results without disclosing the actual sensitive values of individual records.

- **Multilevel Security Models:** Databases can use security levels to classify and restrict data access. Users are only permitted to access data levels for which they have specific authorization.
- **Data Encryption:** Individual data items can be encoded or encrypted so they cannot be read without the proper decryption key. This protects sensitive information even if the raw database is accessed.
- **Randomization Methods:** These methods involve adding "noise" to the data to mask the original attribute values of individual records. The noise must be large enough so that sensitive individual values cannot be recovered.
- **k-anonymity Method:** This technique alters individual records so that any single person cannot be uniquely identified among a group of at least 'k' individuals. It prevents the identification of specific people within a dataset.
- **l-diversity Method:** Similar to k-anonymity, l-diversity ensures that sensitive attributes have enough diversity within a group. This prevents an attacker from making accurate inferences about a person's private data even if they are in an anonymous group.

12. Describe the classification and analysis of Time-Series Data.

Time-series data consists of sequences of numeric data recorded at equal time intervals, which require specific analytical techniques.

- **Trend and Long-Term Movements:** Analysis identifies the general direction in which a time-series graph is moving over a long period. Techniques like weighted moving averages are often used to smooth out short-term fluctuations.

- **Cyclic Movements:** These are long-term oscillations or waves that occur around a primary trend line or curve. They represent broader economic or natural cycles that repeat over several years.
- **Seasonal Variations:** Analysis tracks nearly identical patterns that repeat during specific seasons every year, such as increased retail shopping during holidays. Identifying these helps businesses prepare for predictable annual peaks.
- **Dimensionality Reduction:** For similarity searches, it is often necessary to reduce the complexity of the data using techniques like PCA or Discrete Fourier Transform (DFT). This makes the analysis of long sequences more efficient.
- **Regression Analysis:** This is used to model the relationship between variables over time to project future yields or workloads. It is widely used in stock market analysis and economic forecasting.

13. Explain the role and components of an Indexer in a search engine.

The indexer is the module responsible for organizing web data into a searchable format after it has been fetched by the crawler.

- **Keyword Identification:** The indexer identifies words that describe the main content of a web page. It focuses on specific HTML elements like title tags and heading tags where the most important terms are usually located.
- **Stop Word Removal:** Common, non-descriptive words like "the," "and," and "it" are filtered out during this stage. Removing these "stop words" ensures the index remains focused on meaningful search terms.
- **Stemming Conversion:** The indexer uses stemming to convert various forms of a word into its single root, such as "computers" to "computer". This allows the engine to return relevant results even if the user's exact word form varies.
- **NLP Integration:** Modern indexers use Natural Language Processing (NLP) to go beyond simple keywords and understand the actual context and meaning of web content. This improves the accuracy of the index for complex queries.
- **Keyword-to-ID Mapping:** The final role is to create a comprehensive map that links every identified keyword to the unique IDs of the web pages where it appears. This map is what the query processor uses to find results instantly.

14. Discuss the trends in Big Data Mining and its integration with AI

Modern data mining is evolving to handle increasingly large datasets by integrating with artificial intelligence and cloud technologies.

- **Scalable Mining Methods:** As the volume of data grows rapidly, it is essential to develop algorithms that can process huge amounts of information efficiently and interactively. Scalability is now a primary requirement for all new mining functions.
- **Integration with AI and ML:** Data mining is being combined with machine learning to improve the accuracy of predictions and the discovery of complex patterns. This allows for more autonomous and intelligent data analysis.
- **Cloud-Based Processing:** There is a major trend toward cloud-based data mining, which provides scalable storage and massive processing power for complex datasets. This allows organizations to run mining tasks without expensive local hardware.
- **Real-Time Stream Mining:** Applications like intrusion detection and stock analysis now require models that can analyze data streams in real-time as they flow into the system. This allows for instant detection of anomalies or trends.

- **Visual and Audio Mining:** A recent focus is on integrating data mining with human visual and audio systems to discover knowledge from huge datasets. This trend uses visualization tools to help humans perceive patterns that might be missed by purely automated systems.

2-Mark Questions

15. Define the "Scarcity" and "Abundance" problems in web content mining.

These are common issues encountered when trying to retrieve information from semi-structured web data.

- **Scarcity Problem:** This occurs when a user's query is too specific, resulting in the search engine returning very few or no results.
- **Abundance Problem:** This happens when a query is too general, causing the engine to return an overwhelming number of results that are difficult to filter.

16. What is a Web Crawler, and what are its alternate names?

A web crawler is a fundamental software component of any search engine architecture.

- It is a program that visits every web page available on the World Wide Web to fetch data for later indexing.
- **Alternate Names:** A web crawler is also commonly referred to as a **spider**, a **robot**, or simply a **bot**.

17. Define "Shingles" in the context of document fingerprinting.

Shingles are used as the basic unit of measurement in document similarity analysis.

- A shingle is a continuous sequence of words with a fixed length (e.g., 2 or 3 words) extracted from a document.
- **Purpose:** Similarity between two documents is found by counting how many of these matching word sequences they share.

18. Distinguish the role of "Stop Words" and "Stemming" in indexing.

Both are preprocessing steps used to clean and organize keywords before they are added to a search index.

- **Stop Words:** These are common words like "the" or "and" that are removed because they lack significant meaning for search queries.
- **Stemming:** This is the process of converting various forms of a word (like "computing") back to its root form (like "computer") to ensure better matching.

19. What is a "Damping Factor" in the PageRank formula?

The damping factor, denoted as 'd', is a critical variable in the probability-based PageRank equation.

- It represents the probability that a user will continue clicking on links as they browse.
- **User Exit Role:** The term in the formula specifically represents the scenario where a user chooses to leave the current web page instead of following a link.

20. Define Cyber-Physical Systems (CPS) in the context of data mining.

CPS data mining involves analyzing systems that integrate physical processes with networked information components.

- These systems consist of a large number of interacting physical and information components, such as a patient monitoring system linked to an emergency network.
- **Example:** It is used in transportation systems to link sensor networks and video cameras with traffic control and information centres.

21. What is the difference between a Hub Page and an Authority Page?

These two page types form the basis of link-analysis algorithms like HITS.

- **Authority Page:** This is a page that contains useful, reliable, and important information on a specific topic.
- **Hub Page:** This is a page that acts like a directory by providing links to many different high-quality authority pages.

22. Define Spatial Data Mining.

Spatial data mining discovers knowledge from data related to physical locations and geographic space.

- It is the process of finding patterns in geo-space-related data typically stored in geospatial repositories.
- **Data Types:** This data can be in "vector" formats or in the form of satellite imagery and geo-referenced multimedia.

23. What is Sequence Classification?

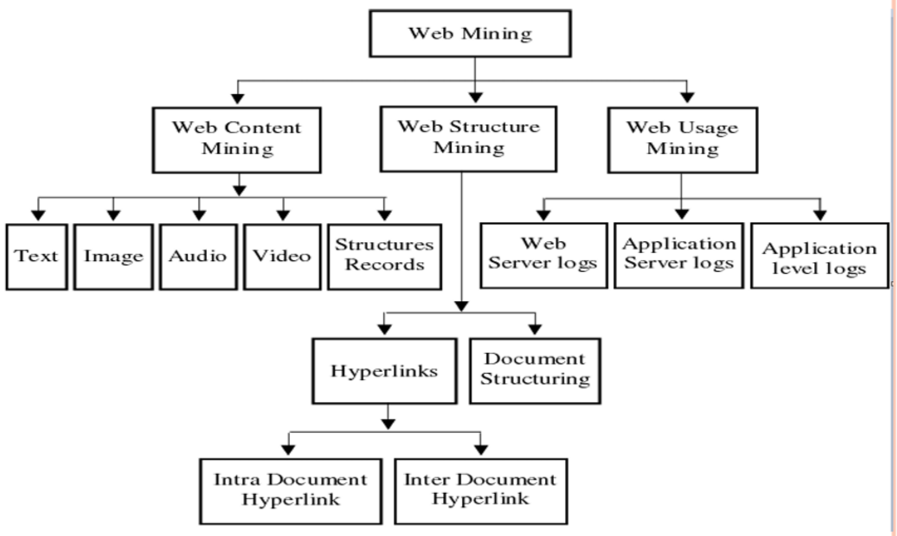
Sequence classification is a mining task that categorizes ordered lists of events into predefined groups.

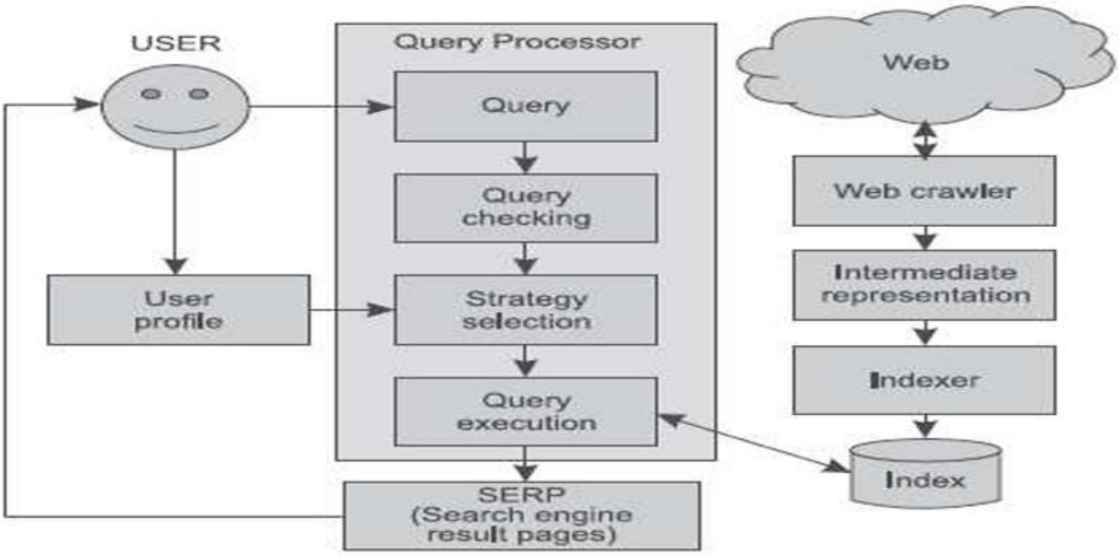
- **Methods:** It can be performed using **feature-based** classification, **sequence distance-based** classification, or **model-based** methods like hidden Markov models.
- **Goal:** The goal is to determine the category of a sequence based on its ordered patterns and semantic meaning.

24. Define Collaborative Filtering.

Collaborative filtering is a popular technique used by recommendation engines like Netflix.

- It is a method that collects information from the interactions of **many different users** to derive suggestions for a specific person.
- **Logic:** It makes recommendations based on identifying other users who have demonstrated similar tastes or were in similar situations.





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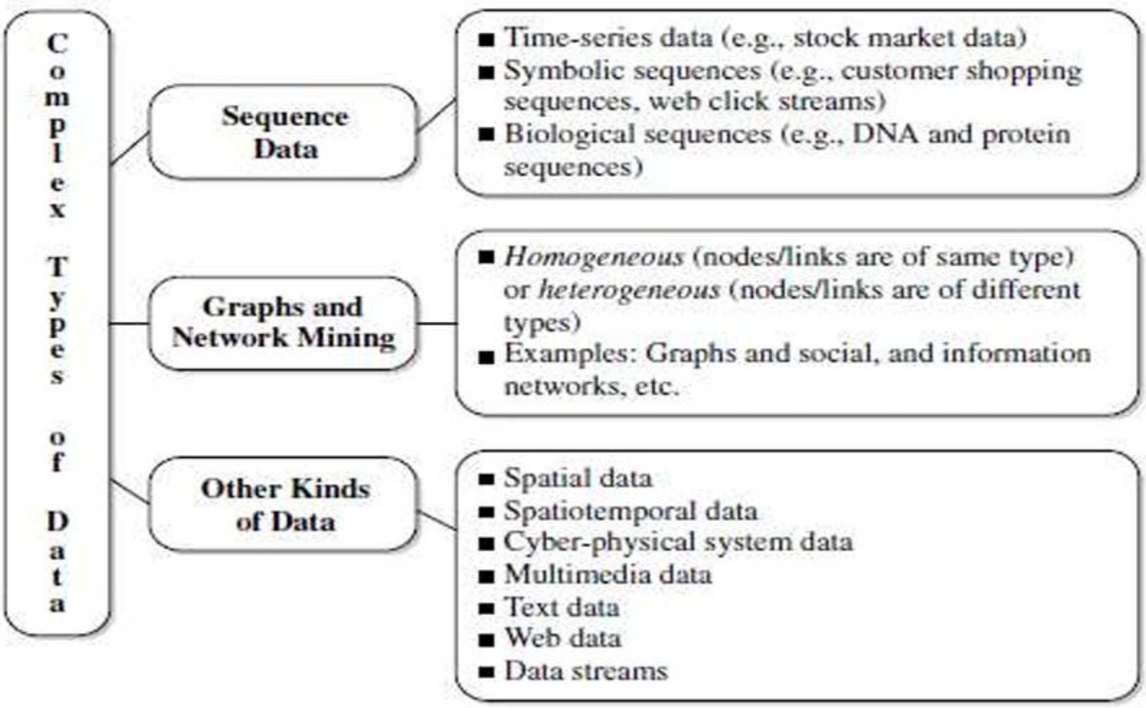

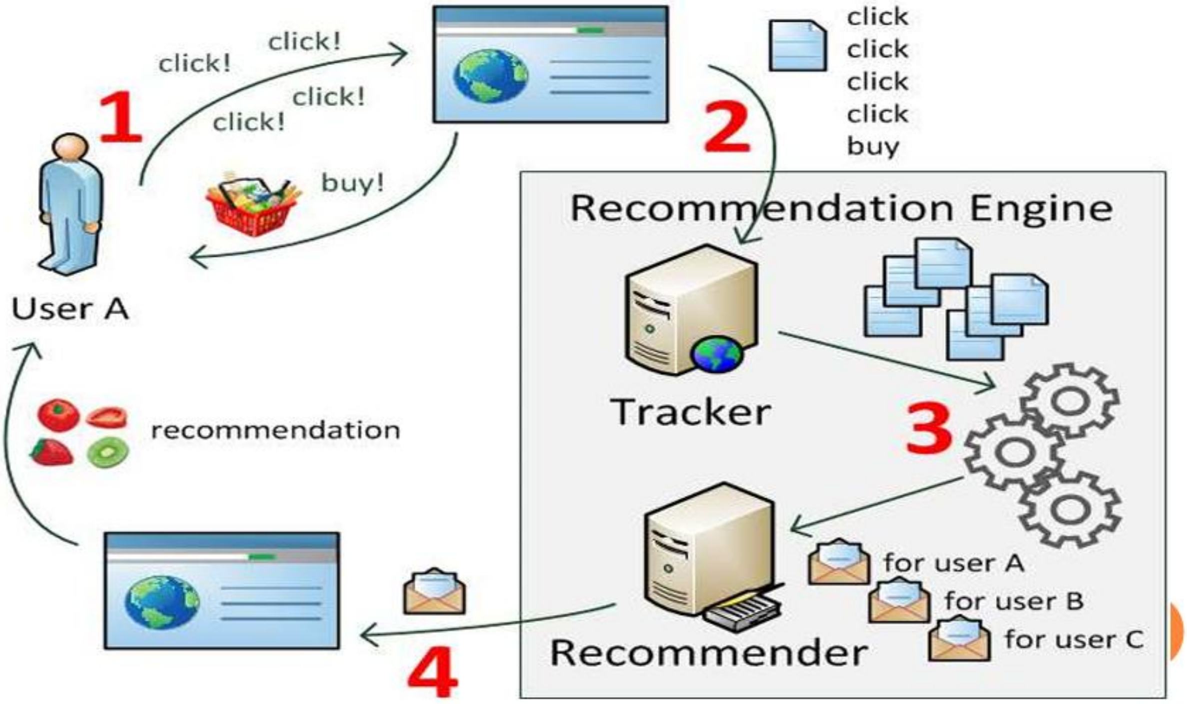
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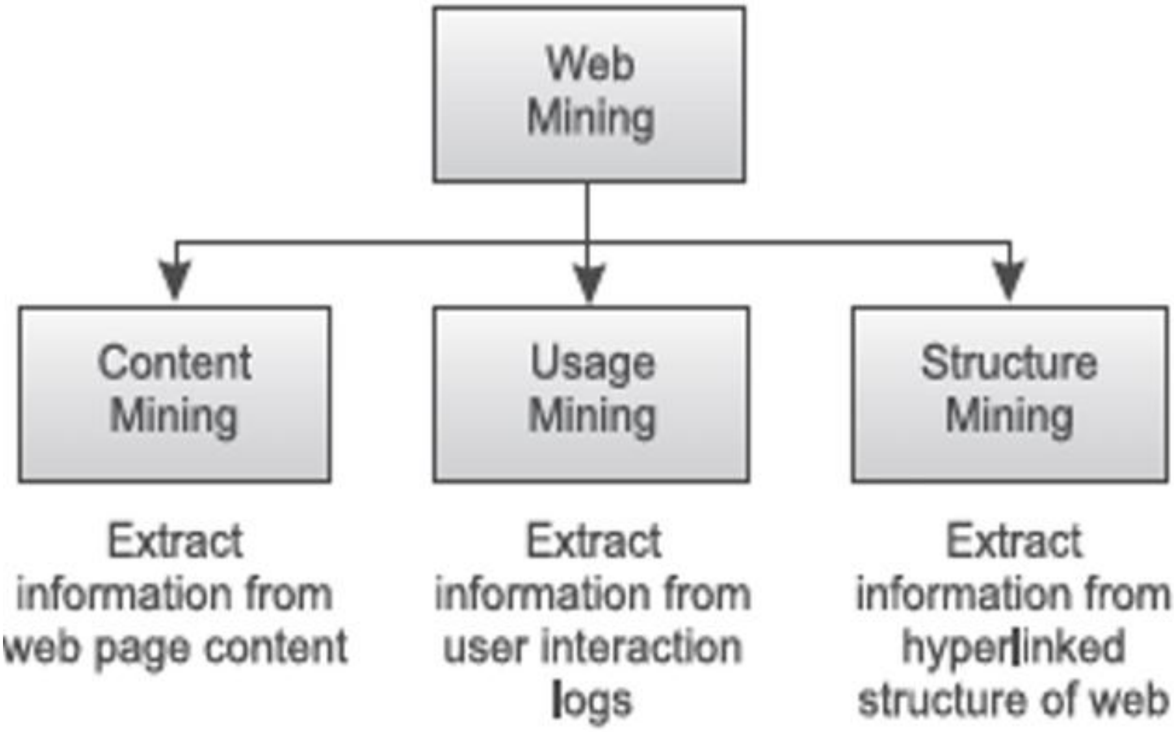
Web Mining
Content Mining
Usage Mining
Structure Mining
Extract information from web page content
Extract information from user interaction logs
Extract information from hyperlinked structure of web